



BEFORE WE BEGIN

Here are some key facts about this program:

- You are working as a Junior Data Scientist at BCG X
- You work within a larger team, where each member has a different role and level of responsibility.
- Your team has been assigned a new project for a client called **PowerCo**
- As a Junior Data Scientist, this is an opportunity to get valuable hands-on experience. You'll work on your programming, problem solving and communication skills - all essential skills for life on the job as a data scientist!

Important note: You'll need to have an understanding of Python before starting this program. If you do not know Python, you are welcome to give it a try, but it might be a little challenging for you!

First, let's find out more on being a Data Scientist at BCG X.



WHAT IS A DATA SCIENTIST?

What exactly is a Data Scientist?

- A Data Scientist works with data to try and deliver value to a business.
- But how is that achieved?
 - Data Science can sometimes overlap with domains such as Data Analysts, Data Engineers and DevOps.
 - How much the Data Scientists role overlaps with these other roles really depends on the company, but in essence a Data Scientist is generally focused on modeling data to be able to accurately predict an outcome, for example, predicting how likely customers are to leave.
 - 4 core skills are used as a Data Scientist: statistics, mathematics, programming and communication.

Now let's find out what Data Science looks like at BCG X.



KEY ROLES AND RESPONSIBILITIES OF A DATA SCIENTIST AT BCG X

BCG X is transforming businesses using data science to help companies generate competitive advantage. To do this, we typically follow a 5-step methodology:

1. **Business understanding & problem framing:** what is the context of this problem and why are they trying to solve it?
2. **Exploratory data analysis & data cleaning:** what data are we working with, what does it look like and how can we make it better?
3. **Feature engineering:** can we enrich this dataset using our own expertise or third party information?
4. **Modeling and evaluation:** can we use this dataset to accurately make predictions? If so, are they reliable?
5. **Insights & Recommendations:** how we can communicate the value of these predictions by explaining them in a way that matters to the business?

The tasks in this program will be focused on using different parts of this methodology at different times, so you'll get a taste of the overall process.

It's a really exciting time to be working with BCG X as more clients are needing data to drive key decisions. So, let's check out what case you'll be working on!



THE BRIEF FROM POWERCO

The Associate Director (AD) of the Data Science team held a team meeting to discuss the client brief. You'll be working closely with Estelle Altazin, a senior data scientist on your team.

Here are the key takeaways from the meeting:

- Your client is **PowerCo** - a major gas and electricity utility that supplies to small and medium sized enterprises.
- The energy market has had a lot of change in recent years and there are more options than ever for customers to choose from.
- PowerCo are concerned about their customers leaving for better offers from other energy providers. When a customer leaves to use another service provider, this is called **churn**.
- This is becoming a big issue for PowerCo and they have engaged BCG to help diagnose the reason why their customers are churning.

During the meeting your AD discussed some potential reasons for this churn, one being how “sensitive” the price is. In other words, how much is price a factor in a customer’s choice to stay with or leave PowerCo?

So, now it’s time for you to investigate this hypothesis.



KNOW WHAT YOU NEED

While you were reviewing the case materials, your team held a team meeting to discuss the client brief and generate hypotheses about the cause.

During the meeting the someone on your team noted that the client strongly believes that the for churn (among others) is that customers have become more “sensitive” to the price and are opting toward low-cost providers – we need to test this hypothesis.

You will partner with Estelle, a senior data scientist, to uncover the insights we need to validate or disprove the idea that price is the critical factor.

You and Estelle know that the first step in any case is to **understand the business and frame the problem** – we need the right data points and techniques to do that.

To ensure we can successfully identify the cause, your task is to draft an email clearly outlining the *data that we’ll need from the client* and the *techniques* you will use to *investigate the issue*.

To help get you started, ask yourself:

1. What are all of the possible reasons a customer would choose to stay with or switch energy providers? (for example: price, energy source, customer service, etc.)
2. What data points would be needed in order to effectively analyze causes of churn? (e.g. purchasing trends, location of business etc.)
3. Once we have the data, what steps will you take to process and analyze the information? (hint: use the 5-step process outlined previously)

Use the text field below to write your email – good luck!



EXAMPLE ANSWER

Great work! Take a look at the example answer below to see how a professional would have attempted this task. Think about what you did well and how you can improve.

Hi [Name],

In order to test the hypothesis that churn is driven by the customers' price sensitivity, we need to model churn probabilities of customers and derive the effect of prices on churn rates.

We would need the following data to be able to build the models:

1. Customer data - which should include characteristics of each client, for example, industry, historical electricity consumption, date joined as customer etc
2. Churn data - which should indicate if customer has churned
3. Historical price data – which should indicate the prices the client charges to each customer for both electricity and gas at granular time intervals

Once we have the data, the work plan would be:

1. Define what price sensitivity is and calculate it
2. Prepare the data and engineer features
3. Test our hypothesis using a binary classification model (e.g. Logistic Regression, Random Forest) to predict likelihood of churn
4. Select the best model based on accuracy metrics and other considerations such as explainability
5. With the trained model, extrapolate the extent to which price sensitivity influences churn

Regards,

[Your name]